Kaushlendra Verma

Post-Doctorate
Meteo-France, Toulouse

Date of Birth: Nov. 10, 1992

Toulouse-31300, France

(+33)-751-53-27-53

er.kaushlendrav@gmail.com

Google Scholar: Kaushlendra Verma

ORCID:0000-0003-4722-8806

Interests

- Hydrological Modelling
- Remote Sensing
- Radar Altimetry
- Machine Learning

Skills

Hydrological Models:

- ISBA-CTRIP
- WRF-Hydro
- SWAT

Data Assimilation:

- CTRIP-HyDAS
- LETKF, Localization

Remote Sensing:

- SWOT, GRACE, and GLDAS

Geospatial Tools:

- ArcGIS, QGIS, and SNAP

Working Knowledge:

- High-Performance Computing
- Google Cloud Computing

Research Objective

To advance regulation-aware, observation-driven river discharge prediction by integrating satellite altimetry, hydrological routing, and ensemble data-assimilation. My long-term aim is to deliver robust, reproducible, and regulation-aware river discharge projections for climate adaptation and water risk management.

Experience

2025 – 2026 **Post-Doctorate (CNRS)** — **Meteo-France, Toulouse**
Title: ESA CCI Discharge Assimilation.
Supervisor: Dr. Simon Munier

2023 – 2025 **Post-Doctorate (CNES)** — **Meteo-France, Toulouse**
Title: Towards a Global Scale SWOT-CTRIP Hydrological Data Assimilation System.
Supervisor: Dr. Simon Munier and Dr. Aaron Boone

Education

2018 – 2022 **Ph.D. in Remote Sensing** — **IIT Bombay, India**
Title: Potential of Surface Water and Ocean Topography (SWOT) Mission for Inland Hydrology.
Supervisor: Dr. J. Indu

2016 – 2018 **M.Tech. in Water Resource Engineering** — **VNIT-Nagpur, India**
Title: Validation of Sensitivity of GRACE and GLDAS Data to Groundwater Variation within Basaltic Aquifer System using Spatial Analysis and ANN.
Supervisor: Dr. Y.B.Katpatal

2010 – 2014 **B.Tech. in Civil Engineering** — **UPTU, India**
Title: Analysis and Design of a Multi-Storey Buildings.
Supervisor: Mr. Shailendra Kumar Prajapati

Training

2023 **Adaptation and development of skills** — **CERFACS**
Title: Training on Data Assimilation by Centre Europeen de Recherche et de Formation Avancee en Calcul Scientifique.

2020 **Community WRF-Hydro Modeling System Abridged Virtual Training** — **NCAR**
Title: First virtual abridged WRF-Hydro Training Workshop by National Center for Atmospheric Research.

Short-term Courses

2016 **Global Initiative of Academic Networks** — **IIT BBS**
Title: Extreme Weather and Climate Variability: Observation, Understanding and Prediction.

2016 **Global Initiative of Academic Networks** — **IIT Madras**
Title: Hydro-informatics for Integrated Water Resource Management using SWAT-Model.

Awards and Achievements

2023 – 2025 Centre National d'Etudes Spatiales (CNES)-Post Doctoral Fellowship 2022.
2021 Awarded AGU Fall meeting 2021 Travel Grant.
2018 – 2023 Awarded MHRD India Fellowship for pursuing Ph.D.
2016 – 2018 Awarded MHRD India Fellowship for pursuing M.Tech.

Metrics

Profiles

Languages

English (Working Language)

French (Basic: A1–A2)

Publications

Journals

- **Verma K.**, and Munier, S. (2026). “Confidence-Aware Framework for Mapping Satellite-Derived River Reaches to Gridded Routing Networks.” Geoscientific Model Development, submitted
- **Verma K.**, Munier, S., Boone, A., and Le Moigne, P. (2026). “CTRIP-HyDAS: A Global Assimilation Framework for SWOT-Derived Discharge Observations.” Water Resources Research, accepted 10.1029/2025WR040888
- Sadki, M., Noual, G., Munier, S., Pedinotti, V., **Verma K.**, Albergel, C., Biancamaria, S., and Andral, A. (2026). “Improving River Routing via ESA CCI Discharge Assimilation.” Hydrology and Earth System Sciences, under review
- Cerbelaud, A., David, C. H., Pavelsky, T. M., Biancamaria, S., . . . **Verma, K.**, et al. (2025). “Satellite Requirements to Capture Water Propagation in Earth’s Rivers.” Reviews of Geophysics, 10.1029/2024RG000871
- **Verma K.** and Indu J. (2023), “Applicability of SWOT data in Calibrating WRF-Hydro Hydrological Model over the Tawa River basin”, Geocarto International, 38(1). 10.1080/10106049.2023.2185292
- **Verma K.,** Nair A., Indu J., Karmakar S. and Calmant S. (2021), “Satellite Altimetry for Indian Reservoirs”, Water Science and Engineering, 14(4),277-285. 10.1016/j.wse.2021.09.001
- **Verma K.** and Indu J. (2021),“Effect of Satellite Altimetry Sampling Error in Estimating Reservoir Storage and Outflow”,Geocarto International. 10106049.2021.1980615
- Nair A. **Verma K.,** Ghosh S., Karmakar S. and Indu J. (2021), “Exploring the Potential of SWOT Mission for Reservoir Monitoring in Mahanadi basin”, Advances in Space Research, 69 (3),1481-1493. 10.1016/j.asr.2021.11.019
- **Verma, K.** and Katpatal, Y. B. (2019),“Groundwater Monitoring Using GRACE and GLDAS Data after Downscaling Within the Basaltic Aquifer System”, Groundwater, 58(1),143–151. 10.1111/gwat.12929

Book Chapters

- **Verma K.** (2026),“From Satellite to Streams: A Review of River Observability in the SWOT Era”, In: Poonia, V., Swarnkar, S., Munoth, P. (eds) Climate Extremes and Emerging Solutions: Data-Driven Insights and Geospatial Techniques. Springer, Cham., 401–415 doi.org/10.1007/978-3-032-14457-7$_2$0
- Indu J., Nair A., Pradhan A., Mangla R., Krishnan S. **Verma K.**, and Huggannavar V. (2022), “Terrestrial water budget through radar remote sensing”, In Earth Observation, Radar Remote Sensing, Elsevier, 123-148. 10.1016/B978-0-12-823457-0.00005-7
- **Verma K.** and Katpatal Y.B. (2021). “Monitoring of Soil Moisture Variability and Establishing the Correlation with Topography by Remotely Sensed GLDAS Data. In: Pandey A., Mishra S., Kansal M., Singh R., Singh V. (eds) Water Management and Water Governance. Water Science and Technology Library, vol 96. Springer, Cham. 10.1007/978-3-030-58051-3-10

International Conferences

- **AGU Fall Meeting** (2021, 2024, 2025)
- **SWOT Science Team Meeting** (2022, 2023, 2024, 2025)
- **Hydrospace** (2023)
- **30 Years of Radar Altimetry** (2024)
- **SWAT India Conference** (2018)

International-Collaboration

- **Members**: NASA-CNES SWOT Science Team (2022 – Present)
- **Contributor**: ESA Climate Change Initiative (CCI) Discharge Project
- **Collaborations**: IRD, INRAE, LEGOS, CERFACS (France), and DST (India)